

LeWAM: a Latent World-Action Model

joint (state, action) prediction — the WAM sibling to LeWM

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notes for discussion with R. Balestriero

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From VLA / BC to World–Action Models (WAM)

- VLA / BC: observation $\rightarrow$ action — imitate, no explicit dynamics.
- **WAM**: jointly model world *and* action (predict next state *and* action) $\Rightarrow$ a forward model you can plan with / verify against, not just imitate.
- Community shift 2024 $\rightarrow$ 26: explicit world dynamics over BC-only (the trend you flagged).
- **LeWAM** = LeWM's JEPA latent + a jointly-trained action pathway. “All-in, all-out” state–action; the *latent* WAM in your Le-family.

The WAM / latent-planner landscape (the fast-moving part)

family	examples (arXiv)	how it gets actions
pure CEM/MPC, no prior	LeWM 2603.19312, DINO-WM 2411.04983, V-JEPA2-AC 2506.09985	sample + score; no learned proposal
policy-prior warm-starts search	TD-MPC2 2310.16828, PiJEPA 2603.25981, Newt 2511.19584	learned π seeds MPPI/CEM
amortize / replace search	GC-IDM 2605.08732 (on LeWM latent!), TD-JEPA 2510.00739	$(z, z_g, h) \rightarrow a$, no search
search-free WAM	DreamZero 2602.15922 (video), Being-H0.7 2605.00078 (latent)	direct action, <i>no</i> test-time search
hierarchical / diffusion-exec	LDP 2504.16925, WorldDP 2606.08775, FF-JEPA 2606.09311	WM picks subgoals, policy executes
"repr. > planning" challenge	MR.Q 2606.05555	model-free + aux predictive losses

Axes to keep in mind: **search** $\leftrightarrow$ **amortize**, **flat** $\leftrightarrow$ **hierarchical**, **pixel/video** $\leftrightarrow$ **latent**.

The WAM field is going *search-free* — LeWAM asks “should we still plan?”

Search-free WAMs (the trend)

- **DreamZero**: video diffusion, $p(\text{video}, a)$, physics from internet-scale video.
- **Being-H0.7**: a *latent* WAM (our exact term!), 200k h egocentric video, future-aware latent workspace.
- Both *amortize planning into one forward pass* — no test-time search / rollout.

LeWAM (contrarian)

- Light JEPA latent (~ 25 GB), joint (z_{t+1}, a_t) .
- *Keeps* a short-horizon search loop + history/past-action policy.
- Asks: *when is search worth keeping?*

Positioning. LeWAM is *not* “the latent WAM” (Being-H0.7 owns that at scale). It is the **per-family analysis** of when latent WM-planning pays off (geometry: yes; contact: no $\Rightarrow$ fall back to history-BC) — the question the search-free WAMs sidestep by never searching.

- Encoder (DINOv2/ViT) $\rightarrow$ latent z ; WM predictor for z -dynamics, SIGReg-regularized $\Rightarrow$ isotropic, plannable.
- Action / intention head: autoregressive on latent history $z_{\leq t}$ + past actions $a_{< t}$ $\rightarrow$ action-embedding $\rightarrow$ decoder $\rightarrow$ raw action.
- Frozen-CLIP task token (multitask); padded+masked action dims (cross-embodiment).
- One latent, three eval modes: history-BC / intuition-guided / CEM-planning.

Finding 1 — planning is *per-family*

- WM-planning (CEM) **works** on geometry / goal-reaching: OGBench-Cube 68 $\rightarrow$ 78 $\rightarrow$ 88, Push-T, Reacher, Two-Room.
- **Collapses to** ≈ 0 on contact-rich robomimic Lift/Can/Square.
- **Mechanism** (verified): both are grasp tasks, identical eval — so *not* “push vs grasp.” On Cube the goal-image shows the object *displaced* $\Rightarrow$ CEM must grasp+carry to lower latent cost. On Lift the goal is the cube +few cm, *swamped by arm pose* $\Rightarrow$ CEM matches pose, skips the grasp; grasp only weakly exposed in one frame (probe 0.82 $\rightarrow$ 0.99 with history). On **Can/Square** the latent *does* see grasp (0.96/0.999) but the task is *multi-stage* (approach $\rightarrow$ grasp $\rightarrow$ place) so flat CEM still fails. (Data: cube $\sim$ 1M vs robomimic 200 demos.)

- On robomimic, history / past-action conditioned BC **recovers grasp state**: post-grasp $P(\text{success} | \text{grasp})$ $76 \rightarrow 99$.
- history-BC: Lift ~ 95 , Can $\sim 48\text{--}72$, Square $\sim 60\text{s}$ vs online CEM = 0.
- The action pathway, conditioned on *past actions*, carries contact tasks where single-frame WM planning cannot.
- Closest published latent planner LDP 0.69/0.70/0.46; we beat it on Square.

Finding 3 (honest negative) — the action-head A/B is within noise

- detach=TRUE / FALSE / solveact (raw action) / SIGReg-on-act-emb: all **tied** $\approx 72 \pm 5$ **on Can**, $N=50$, 3 *training* seeds.
- The ~ 10 -pt **training-run variance swamps** the 4–7-pt effects single runs showed.
- Lesson: multi-*train*-seed is mandatory; single-run A/Bs (and eval-seed-only “3-seed”) mislead. We caught a false “solveact wins / intent_loss hurts” headline before publishing.

- LeWAM = unified JEPA *latent* world-action model. Contributions:
 - ① the per-family “*when does WM-planning help*” analysis;
 - ② history / past-action-conditioned action pathway recovering contact state (history-BC $\gg$ CEM);
 - ③ the WAM reframe + honest variance accounting (the latent counterpart to DreamZero’s video WAM).
- **Not** a fine action-head SR win — that washed out. The story is mechanism + analysis + the history-conditioned wedge.

- **DreamZero** (video WAM, no search): LeWAM = light latent + the planning analysis it sidesteps.
- **Newt** / **PiJEPA** (JEPA priors, Markovian/raw-action): LeWAM = history+past-action conditioned.
- **GC-IDM** (amortize on LeWM latent — must-beat): LeWAM = warm-start-*then*-refine, not *replace* search.
- **WorldDP** / **LDP** (hierarchical + diffusion exec): they *assume* flat planning fails; we *map when* it fails.
- **MR.Q** (repr. > planning): our action pathway *is* that predictive representation; we restrict planning to where it provably helps.

- **Hierarchical subgoal-LeWAM** (WorldDP / LDP style): WM plans subgoal latents, the history policy executes each $\Rightarrow$ attack robomimic planning = 0 (the per-family limit). Implemented EMA-target + the scaffolding; test next.
- Unified “z, a interleaved, all-in-all-out” tokenization (one trunk).
- Open test: does subgoaling beat flat history-BC where a single frame only weakly exposes contact?

- Framing & venue: LeWAM as the latent-WAM showcase (your “different needs than pure LeWM”).
- Which to push first: the hierarchical-subgoal method, or the per-family planning *analysis* paper?
- Your read on positioning vs the video-WAM wave (DreamZero) — is the per-family planning story the right headline?

(numbers to re-pull fresh before presenting: robomimic histbc 3-seed, cube 68/78/88, the Can A/B 74/72/76 $\approx$ 70/70 = tied.)